Faculty of Information Technology
20241

ASPU
Al-Sham Private University

Multimedia-Lecture-Seven

Data Compression

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Multimedia-Lecture-Seven

Data Compression

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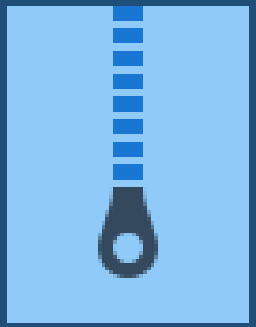
Data Compression Concept



Most of the real data is very redundant. Data Compression is basically defined as a technique that decreases the size of data.

It reduces the storage space and hence storage cost, also reduces time to retrieve and transmit data.

Data-compression techniques can be divided into two major families:
- Lossy Data Compression.
- Lossless Data Compression.

Lossy and Lossless Compression

Original Image

Lossless Compression

Original Lena Image (12KB size)

Lena Image, Compressed (85% less information, 1.8KB)

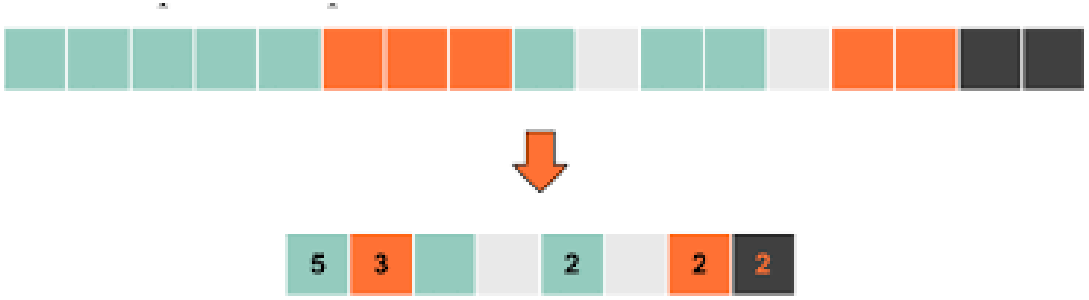
Lena Image, Highly Compressed (96% less information, 0.56KB)

Data Compression with PYTHON

- Compression Using RLE Method
- De-compression Using RLE Method
- Image Compression Using RLE Method

Encoding Data Using RLE Method

Run Length Encoding (RLE): is a simple compression algorithm used to compress sequences containing subsequent repetitions of the same character.

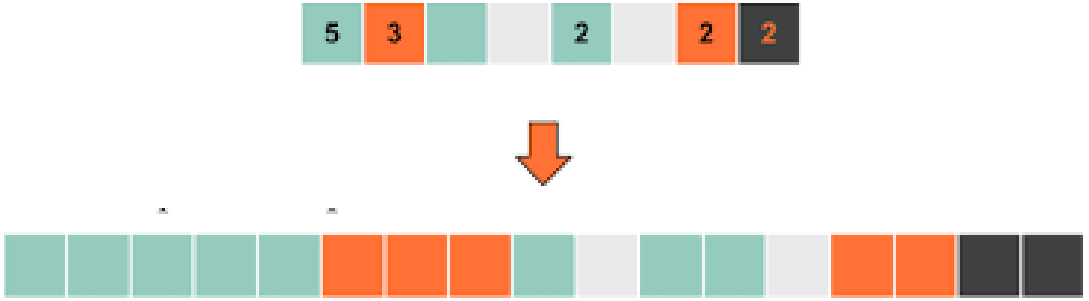


Example:

```
%% input data
x = [5, 5, 2, 1, 1, 1, 1, 3]
y = []
c = 1

for i in range(len(x) - 1):
    if x[i] == x[i + 1]:
        c += 1
    else:
        y.extend([c, x[i]])
        c = 1

y.append(c)
y.append(x[-1])
print(y)
```

What is The Output??

Decoding Data Using RLE Method



Example:

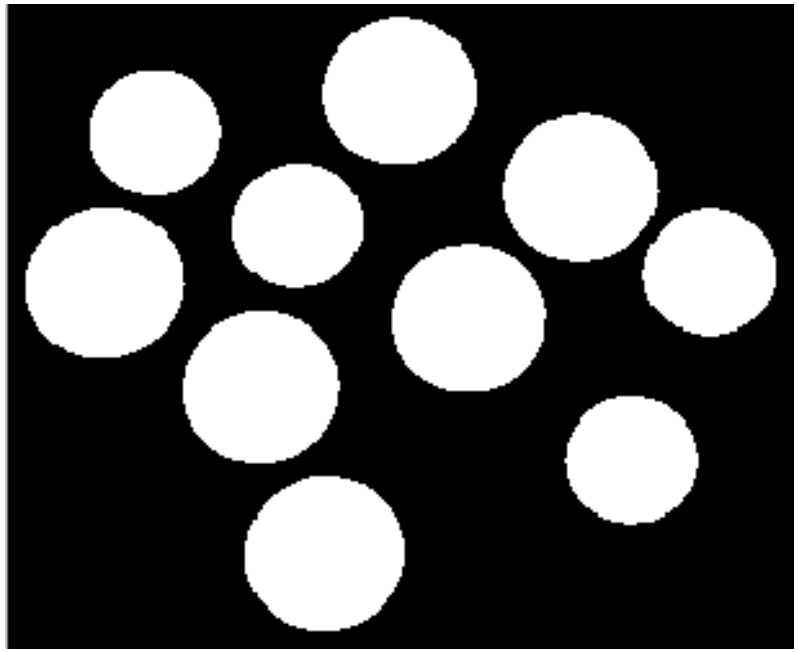
```
%% input data
y = [5, 5, 2, 1, 1, 1, 1, 3]
decoded_data = []

for i in range(0, len(y), 2):
    count = y[i]
    value = y[i + 1]
    decoded_data.extend([value] * count)

print("Decoded Data:", decoded_data)
```

What is The Output??

Image Compression Using RLE Method

Try the previous code for an black and white image blobs.png.

Tip:
Use this code for comparing elements:
np.array_equal(x[i], x[i + 1])



That’s All